



### How to choose forecasting models

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## Data transformations and forecasting models: what to use and when

| Transformation                          | Properties                                                                                                                                                                                                                                                                        | When to use                                                                                                                                                                                                                                                                                              | Points to keep in mind                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deflation by CPI or another price index | Converts data from nominal dollars (or other currency) to constant dollars; usually helps to stabilize variance                                                                                                                                                                   | When data are measured in nominal dollars (or other currency) and you want to explicitly show the effect of inflation--i.e., uncover "real growth"                                                                                                                                                       | To generate a true forecast for the future in nominal terms, you will need to make an explicit forecast of the future value of the price index--i.e., you will need to forecast the inflation rate (but this is easy if you're in a period of steady inflation)                                                                                                                      |
| Deflation at a fixed rate               | Merely applies a constant discount factor to past data                                                                                                                                                                                                                            | When you only need to approximately model the effect of past inflation and/or you wish to impose an assumption about the current and future inflation rate--you can twiddle the inflation rate to see what value does the best job of flattening out the trend and/or stabilizing the variance           | When used with a zero-trend model like simple exponential smoothing or random walk without growth, the assumed inflation rate is precisely the percentage growth in the future forecasts.                                                                                                                                                                                            |
| Logarithm                               | Converts <i>multiplicative</i> patterns to <i>additive</i> patterns and/or linearizes exponential growth; converts <i>absolute</i> changes to <i>percentage</i> changes; often stabilizes the variance of data with compound growth, regardless of whether deflation is also used | When compound growth is not due to inflation (e.g. when data is not measured in currency); when you do not need to separate inflation from real growth; when data distribution is positive and highly skewed (e.g., exponential or log-normal distribution); when variables are multiplicatively related | Logging is <i>not the same</i> as deflating: it linearizes growth but does not remove a general upward trend; if logged data still have a consistent upward trend, then you should use a model that includes a trend factor (e.g., random walk with growth, ARIMA, linear exponential smoothing).                                                                                    |
| First difference                        | Converts "levels" to "changes"                                                                                                                                                                                                                                                    | When you need to stationarize a series with a strong trend and/or random-walk behavior (often useful when fitting regression models to time series data)                                                                                                                                                 | Differencing is an explicit option in ARIMA modeling and it is implicitly a part of random walk and exponential smoothing models; therefore you would <i>not</i> manually difference the input variable (using the DIFF function) when specifying model type as "random walk" or "exponential smoothing" or "ARIMA"; first difference of LOG(Y) is the <i>percentage</i> change in Y |
| Seasonal difference                     | Converts "levels" to "seasonal changes"                                                                                                                                                                                                                                           | When you need to remove the gross features of seasonality from a strongly seasonal series without going to the trouble of estimating seasonal indices                                                                                                                                                    | Seasonal differencing is an explicit option in ARIMA modeling; you <b>MUST</b> include a seasonal difference (as a modeling option, <i>not</i> an SDIFF transformation of the input variable) if the seasonal pattern is consistent and you wish it to be maintained in long-term forecasts                                                                                          |
| Seasonal adjustment                     | Removes a constant seasonal pattern from a series (either multiplicative or additive)                                                                                                                                                                                             | When you wish to separate out the seasonal component of a series and then fit what's left with a nonseasonal model (regression, smoothing, use the multiplicative version unless data has                                                                                                                | Adds a lot of parameters to the model--one for each season of the year. (In Statgraphics, the seasonal indices are not explicitly shown in the output of the Forecasting procedure--you must separately run the Descriptive Methods procedure to display the seasonal indices.)                                                                                                      |

been logged

| Model type                                       | Properties                                                                                                                                                                                                        | When to use                                                                                                                                                                                                                                                                               | Points to keep in mind                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random walk                                      | Predicts that "next period equals this period" (perhaps plus a constant); a.k.a. ARIMA(0,1,0) model                                                                                                               | As a baseline against which to compare more elaborate models; when applied to logged data, it is a "geometric" random walk--the default model for stock market data                                                                                                                       | Plot of forecasts looks exactly like a plot of the data, except lagged by one period (and shifted slightly up or down if a growth term is included); long term forecasts follow a straight line (horizontal if no growth term is included); confidence intervals for long-term forecasts widen according to a square-root law (sideways-parabola shape); logically equivalent to MEAN model fitted to DIFF(Y)                                                                                                                                           |
| Linear trend                                     | Regression of Y on the time index                                                                                                                                                                                 | Rarely the best model for forecasting--use only when you have very few data points and no obvious pattern in data other than a slight trend; can be used in conjunction with seasonal adjustment--but if you have enough data to seasonally adjust, you probably should use another model | Forecasts follow a straight line whose slope equals the average slope over the whole estimation period but whose intercept is anchored in the distant past; short-term forecasts therefore may miss badly and confidence intervals for long-term forecasts are usually not reliable; other models that extrapolate a linear trend into the future (random walk with growth, linear exponential smoothing, ARIMA models with 1 difference w/constant or 2 differences w/o constant) often do a better job by "reanchoring" the trend line on recent data |
| Simple moving average                            | Simple (equally weighted) average of recent data                                                                                                                                                                  | When data are in short supply and/or highly irregular                                                                                                                                                                                                                                     | Primitive but relatively robust against outliers and messy data; long-term forecasts are a horizontal line extrapolated from the most recent average; a long-term trend can be incorporated via fixed-rate deflation at an assumed interest rate                                                                                                                                                                                                                                                                                                        |
| Simple exponential smoothing                     | Exponentially weighted average of recent data; "average age" of data in forecast (amount by which forecasts lag behind turning points) is $1/\alpha$ ; same as an ARIMA(0,1,1) model without constant             | When data are nonseasonal (or deseasonalized) and display a time-varying mean without a consistent trend                                                                                                                                                                                  | Long-term forecasts are a horizontal line extrapolated from the most recent smoothed value; same as a random walk model without growth if $\alpha=0.9999$ ; forecasts get smoother and slower to respond to turning points as $\alpha$ approaches zero; confidence intervals widen less rapidly than in the random walk model; a long-term trend can be incorporated via fixed-rate deflation at an assumed interest rate or by fitting an ARIMA(0,1,1) model <i>with</i> constant                                                                      |
| Linear exponential smoothing (Brown's or Holt's) | Assumes a time-varying linear trend as well as a time-varying level (Brown's uses 1 parameter, Holt's uses separate smoothing parameters for level and trend); essentially an ARIMA(0,2,2) model without constant | When data are nonseasonal (or deseasonalized) and display time-varying local trends (usually applicable to data that are "smoother" in appearance--i.e., less noisy--than what would be well fitted by simple exponential smoothing)                                                      | Long-term forecasts follow a straight line whose slope is the estimated local trend at the end of the series; confidence intervals for long-term forecasts widen rapidly--the model assumes that the future is VERY uncertain because of time-varying trends; often does not outperform simple exponential smoothing, even for data with trends, because extrapolation of time-varying trends is risky                                                                                                                                                  |
| Seasonal random walk                             | Predicts that "next period equals same period last year" (plus constant); an ARIMA(0,0,0)x(0,1,0) model with constant                                                                                             | As a baseline against which to compare fancier seasonal models; as foundation for seasonal ARIMA models (e.g., (1,0,0)x(0,1,1))                                                                                                                                                           | Long-term forecasts have same seasonal pattern as last year; long-term trend is equal to the <i>average</i> trend over whole past history of series; confidence intervals widen slowly; slow to respond to cyclical upturns and downturns; logically equivalent to MEAN model fitted to SDIFF(Y,s)                                                                                                                                                                                                                                                      |
| Seasonal random trend                            | Predicts that change from this period to next period will be the same as change observed at this time last year; an ARIMA(0,1,0)x(0,1,0) model without constant                                                   | As a baseline against which to compare fancier seasonal models; as foundation for seasonal ARIMA models (e.g., (0,1,1)x(0,1,1) without constant)                                                                                                                                          | Long-term forecasts have same seasonal pattern as last year; long-term trend is equal to the <i>most recently observed</i> annual trend; confidence intervals widen rapidly; quick to respond to cyclical upturns and downturns; logically equivalent to MEAN model fitted to DIFF(SDIFF(Y)) (with no constant--i.e., mean is assumed to be zero)                                                                                                                                                                                                       |
| Winter's seasonal smoothing                      | Assumes time-varying level, trend, and seasonal indices                                                                                                                                                           | When data are trended and seasonal and you wish to decompose it into <i>local</i>                                                                                                                                                                                                         | Initialization of seasonal indices and joint estimation of three smoothing parameters is sometimes tricky--watch to see that                                                                                                                                                                                                                                                                                                                                                                                                                            |

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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | (either multiplicative or additive seasonality)                                                                                                                                                                                                                                     | level/trend/seasonal factors; normally you use the multiplicative version unless data is logged                                                                                                                                                                                                                                                                                 | parameter estimates converge and that forecasts and confidence intervals look reasonable; a popular choice for "automatic" forecasting because it does a little of everything, but has a lot of parameters and sometimes overfits the data or is unstable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Multiple regression | A general linear forecasting equation involving other variables                                                                                                                                                                                                                     | When data are correlated with other explanatory or causal variables (e.g., price, advertising, promotions, interest rates, indicators of general economic activity, etc.); the key is to choose the right variables and the right <i>transformations</i> of those variables to justify the assumption of a linear model and to take into account the time dimension in the data | Forecasts cannot be extrapolated into the future unless and until values are available for the independent variables; for this reason the independent variables must often be lagged by one or more periods--but when <i>only</i> lagged variables are used, a regression model may fail to outperform a time series model which relies only on the history of the original series; regressions of <i>nonstationary</i> variables often have high "R-squared" but poor performance compared to time series models; it often helps to stationarize the dependent variable and/or add lags of the dependent and independent variables to the model; "automatic" model selection techniques such as stepwise regression and all-possible regressions are available, but beware of overfitting; it is important to validate the model by testing it on hold-out data and by computing its "effective R-squared" (percent of variance explained) relative to a random walk model or other appropriate time series model |
| ARIMA               | A general class of models that includes random walk, random trend, seasonal and non-seasonal exponential smoothing, and autoregressive models; forecasts for the stationarized dependent variable are a linear function of lags of the dependent variable and/or lags of the errors | When data are relatively plentiful (4 seasons or more) and can be satisfactorily stationarized by differencing and other mathematical transformations; when it is not necessary to explicitly separate out the seasonal component (if any) in the form of seasonal indices                                                                                                      | ARIMA models are designed to squeeze all autocorrelation out of the original time series; a systematic procedure exists for identifying the best ARIMA model for any given time series; features of ARIMA and multiple regression models can be combined in a natural way; ARIMA models often provide a good fit to highly aggregated, highly plentiful data; they may perform relatively less well on disaggregated, irregular, and/or sparse data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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